Discrete Mathematics

Lecture-3

Boolean Algebra

Definitions:

Let B be a nonempty set with two binary operations $+$ and $*$, a unary operation $'$, and two distinct elements 0 and 1. Then B is called a Boolean algebra if the following axioms hold where a, b, c are any elements in B:

$[B_1]$ Commutative laws:

$$(1a)\ a + b = b + a \qquad (1b)\ a * b = b * a\ [B_2]$$

Distributive laws:

$$(2a)\ a + (b * c) = (a + b) * (a + c) \qquad (2b)\ a * (b + c) = (a * b) + (a * c)$$

$[B_3]$ Identity laws:

$$(3a)\ a + 0 = a \qquad (3b)\ a * 1 = a$$

$[B_4]$ Complement laws:

$$(4a)\ a + a' = 0 \qquad (4b)\ a * a' = 1$$

The operations $+$, $*$, and $'$ are called sum, product, and complement, respectively. We adopt the usual convention that, unless we are guided by parentheses, $'$ has precedence over $*$, and $*$ has precedence over $+$.

For example,

$a + b * c$ means $a + (b * c)$ and not $(a + b) * c$; $a * b'$ means $a * (b)'$ and not $(a * b)'$.

Duality:

The dual of any statement in a Boolean algebra B is the statement obtained by interchanging the operations $+$ and $*$, and interchanging their identity elements 0 and 1 in the original statement. For example, the dual of

$(1 + a) * (b + 0) = b$ is $(0 * a) + (b * 1) = b$

Observe the symmetry in the axioms of a Boolean algebra B. That is, the dual of the set of axioms of B is the same as the original set of axioms. Accordingly, the important principle of duality holds in B.

Theorem: Let a, b, c be any elements in a Boolean algebra B.

(i) Idempotent laws:

$$(1)\ a + a = a \qquad (2)\ a * a = a$$

(ii) Boundedness laws:

$$(1)\ a + 1 = 1 \qquad (2)\ a * 0 = 0$$

(iii) Absorption laws:

$$(1)\ a + (a * b) = a \quad (2)\ a * (a + b) = a$$

(iv) Associative laws:

$$(1)\ (a + b) + c = a + (b + c) \qquad (2)\ (a * b) * c = a * (b * c)$$

Proof: (i)

(1)

$$\begin{aligned} a + a &= (a + a) * 1 && \text{[By Identity law: 3(b)]} \\ &= (a + a) * (a + a') && \text{[By Complement law: 4(a)]} \\ &= a + (a * a') && \text{[By Distributive law: 2(a)]} \\ &= a + 0 && \text{[By Complement law: 4(b)]} \\ &= a && \text{[By Identity law: 3(a)]} \end{aligned}$$

(2)

$$\begin{aligned} a &= a * 1 && \text{[By Identity law: 3(b)]} \\ &= a * (a + a') && \text{[By Complement law: 4(a)]} \\ &= (a * a) + (a * a') && \text{[By Distributive law: 2(b)]} \\ &= (a * a) + 0 && \text{[By Complement law: 4(b)]} \\ &= a * a && \text{[By Identity law: 3(a)]} \end{aligned}$$

(ii)

(1)

$$\begin{aligned} a + 1 &= 1 * (a + 1) && \text{[By Identity law: 3(b)]} \\ &= (a + a') * (a + 1) && \text{[By Complement law: 4(a)]} \\ &= a + (a' * 1) && \text{[By Distributive law: 2(a)]} \\ &= a + a' && \text{[By Identity law: 3(a)]} \\ &= 1 && \text{[By Complement law: 4(a)]} \end{aligned}$$

(2)

$$\begin{aligned} a * 0 &= (a * 0) + 0 && \text{[By Identity law: 3(a)]} \\ &= (a * 0) + (a * a') && \text{[By Complement law: 4(b)]} \\ &= a * (0 + a') && \text{[By Distributive law: 2(b)]} \\ &= a * (a' + 0) && \text{[By Commutative law: 1(a)]} \\ &= a * a' && \text{[By Identity law: 3(a)]} \\ &= 0 && \text{[By Complement law: 4(b)]} \end{aligned}$$

(iii)

(1)

$$\begin{aligned} a + (a * b) &= (a * 1) + (a * b) && \text{[By Identity law: 3(b)]} \\ &= a * (1 + b) && \text{[By Distributive law: 2(a)]} \end{aligned}$$

$= a * (b + 1)$ [By Commutative law: 1(a)]

$= a * 1$ [By Boundedness law]

$= a$ [By Identity law: 3(b)]

$a * (a + b) = (a + 0) * (a + b)$ [By Identity law: 3(a)]

$= a + (0 * b)$ [By Distributive law: 2(b)]

$= a + (b * 0)$ [By Commutative law: 1(b)]

$= a + 0$ [By Boundedness law]

$= a$ [By Identity law: 3(a)]

(iv)

(1) R.H.S $= a + (b + c)$

$= [a + (b + c)] * 1$ [By Identity law]

$= [a + (b + c)] * (c + c')$ [By Complement law]

$= \{[a + (b + c)] * c\} + \{[a + (b + c)] * c'\}$ [By distributive law]

$= \{c * [a + (b + c)]\} + \{c' * [a + (b + c)]\}$ [By commutative law]

$= \{c * a + c * (b + c)\} + \{c' * a + c' * (b + c)\}$ [By distributive law]

$= \{c * a + c\} + \{c' * a + c' * (b + c)\}$ [By absorption law]

$= c + \{c' * a + c' * (b + c)\}$ [By absorption law]

$= c + \{c' * a + c' * b + c' * c\}$ [By distributive law]

$= c + \{c' * a + c' * b + c * c'\}$ [By commutative law]

$= c + \{c' * a + c' * b + 0\}$ [By complement law]

$= c + \{c' * a + c' * b\}$ [By identity law]

$= c + \{c' * (a + b)\}$ [By distributive law]

$= (c + c') * [c + (a + b)]$ [By distributive law]

$= 1 * [c + (a + b)]$ [By complement law]

$= [c + (a + b)] * 1$ [By commutative law]

$= c + (a + b)$ [By identity law]

$= (a + b) + c$ [By distributive law]

$=$ L.H.S

Hence, $(a + b) + c = a + (b + c)$.

(2) R.H.S $= a * (b * c)$

$= [a * (b * c)] + 0$ [By identity law]

$= [a * (b * c)] + (c * c')$ [By complement law]

$= \{[a * (b * c)] + c\} * \{[a * (b * c)] + c'\}$ [By distributive law]

$= \{c + [a * (b * c)]\} * \{c' + [a * (b * c)]\}$ [By commutative law] $=$

$\{(c + a) * [c + (b * c)\} * \{(c' + a) * [c' + (b * c)]\}$ [By distributive law]

$= \{(c + a) * c\} * \{(c' + a) * [c' + (b * c)]\}$ [By absorption law]
$= \{c * (c + a)\} * \{(c' + a) * [c' + (b * c)]\}$ [By commutative law]
$= c * \{(c' + a) * [c' + (b * c)]\}$ [By absorption law]
$= c * \{(c' + a) * [(c' + b) * (c' + c)]\}$ [By distributive law]
$= c * \{(c' + a) * [(c' + b) * (c + c')]\}$ [By commutative law]
$= c * \{(c' + a) * [(c' + b) * 1]\}$ [By complement law]
$= c * \{(c' + a) * (c' + b)\}$ [By identity law]
$= c * \{c' + (a * b)\}$ [By distributive law]
$= c * c' + c * (a * b)$ [By distributive law]
$= 0 + c * (a * b)$ [By complement law]
$= c * (a * b) + 0$ [By commutative law]
$= c * (a * b)$ [By identity law]
$= (a * b) * c$ [By commutative law]
$=$ L.H.S

Hence, $(a * b) * c = a * (b * c)$.

Theorem: Let a be any element of a Boolean algebra B.

(i) (Uniqueness of Complement) If $a + x = 1$ and $a * x = 0$, then $x = a'$.

(ii) (Involution law) $(a')' = a$.

(iii) (a) $0' = 1$. (b) $1' = 0$.

Proof:

(i) We have:

$a' = a' + 0$ [Using identity law]
$= a' + (a * x)$ [$\because a * x = 0$]
$= (a' + a) * (a' + x)$ [By distributive law]
$= 1 * (a' + x)$ [By complement law]
$= (a' + x) * 1$ [By commutative law]
$= (a' + x)$ [By identity law]

Also, $x = x + 0$ [Using identity law]
$= x + (a * a')$ [Using complement law]
$= (x + a) * (x + a')$ [Using distributive law]
$= (a + x) * (a' + x)$ [Using commutative law]
$= 1 * (a' + x)$ [$\because a + x = 1$]
$= a' + x$ [Using identity law]

Hence, $x = a'$.

(ii) Consider $(a')' = (a')' + 0$ [Using identity law]

$$= (a')' + a * a' \quad \text{[Using complement law]}$$
$$= ((a')' + a) * ((a')' + a') \quad \text{[Using distributive law]}$$
$$= ((a')' + a) * 1 \quad \text{[Using complement law]}$$
$$= ((a')' + a) * (a + a') \quad \text{[Using complement law]}$$
$$= a + ((a')' * a') \quad \text{[Using distributive law]}$$
$$= a + (a' * (a')') \quad \text{[Using commutative law]}$$
$$= a + 0 \quad \text{[Using complement law]}$$
$$= a \quad \text{[Using identity law]}$$

(iii) By boundedness law, $0 + 1 = 1$ By identity law, $0 * 1 = 0$

Thus, by uniqueness of complement,

$$1 = 0'$$
$$\therefore 0' = 1$$

By duality, $1' = 0$.

Theorem: (DeMorgan's laws)

(a) $(a + b)' = a' * b'$, (b) $(a * b)' = a' + b'$

Proof:

(a) We need to show that $(a + b) + (a' * b') = 1$ and $(a + b) * (a' * b') = 0$, then by uniqueness of complement, $(a' * b') = (a + b)'$.

We have

$$\begin{aligned}(a + b) + (a' * b') &= b + a + (a' * b') \\ &= b + (a + a') * (a + b') \\ &= b + 1 * (a + b') \\ &= b + a + b' \\ &= b + b' + a \\ &= 1 + a \\ &= 1\end{aligned}$$

Also,

$$\begin{aligned}(a + b) * (a' * b') &= \{(a + b) * a'\} * b' \\ &= \{(a * a') + (b * a')\} * b' \\ &= \{0 + (b * a')\} * b' \\ &= (b * a') * b' \\ &= (b * b') * a' \\ &= 0 * a' \\ &= 0\end{aligned}$$

Thus $a' * b' = (a + b)'$.

(b) By principle of duality, we have, $a' + b' = (a * b)'$.